



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

School of
Professional and
Continuing
Education
(SPACE)

Assignment 1

Course: DSPD 1573 (Programming Fundamentals)

Lecturer's Name: Madam Rokiah binti Bahari

No.	Name	Matric Number
1.	AMIR LUQMAN BIN MOHD KHARIZUL	A26DW0486
2.	GAN ZI XUAN	A26DW1713
3.	NUR FARZANA ARIFAH BINTI MOHD FAUZI	A26DW2175
4.	MUHAMMAD AL HADI BIN MOHD NOR	A26DW1876
5.	BRYENT A/L S.LORENLE	A26DW2215

Problem Analysis

The objective of the **Parking Management System** is to calculate the correct parking fee for a vehicle based on its **entry time, exit time, vehicle type, and membership status**. The system determines the total parking duration, selects the appropriate hourly rate according to the vehicle type, calculates the basic parking charge, applies a membership discount when applicable, and displays the final amount that the customer needs to pay.

The system requires four main inputs: **Entry Time, Exit Time, Vehicle Type, and Membership Status**. From these inputs, it calculates the parking duration, rounds partial hours upward, determines the hourly rate, applies the membership discount, and produces the **Parking Duration, Basic Charge, Discount, and Final Charge** as outputs.

The conditions and rules that need to be considered are:

- The parking duration is calculated using Exit Time – Entry Time
- If the calculated duration is negative, it means the vehicle exited on the following day, so **24 hours are added** to the duration.
- Any partial parking hour is **rounded upward to the next full hour**.
- The hourly rate depends on the vehicle type.
- A **Car** is charged **RM3.00 per hour**.
- A **Motorcycle** is charged **RM1.50 per hour**.
- A **Van** is charged **RM5.00 per hour**.
- If the vehicle type is not Car, Motorcycle, or Van, the system displays **"Invalid vehicle type"** and stops.
- Members receive a **5% discount** on the basic parking charge.
- Non-members receive **no discount**.
- The basic charge is calculated as Parking Hours x Hourly Rates
- The final charge is calculated as Basic Charge – Discount.

Assumptions and Constraints

Assumptions/Constraints	Description
Time Format	Entry and exit times are entered using a 24-hour format . The test cases also show decimal values such as 6.5 to represent 6:30
Parking Period	The system assumes a parking session does not exceed 24 hours , because an overnight session is handled by adding 24 when the exit time is earlier than the entry time.
Partial Hours	Any partial hour is rounded up using the CEILING function. For example, 3.5 hours is charged as 4 hours.
Vehicle Type	Only Car, Motorcycle, and Van are accepted. Other vehicle types are considered invalid.
Car Rate	RM3.00 per hour
Motorcycle Rate	RM1.50 per hour
Van Rate	RM5.00 per hour
Membership Discount	Members receive a 5% discount on the basic charge.
Non-Members	Non-members receive no discount.
Basic Charge	Parking Hours x Hourly Rate
Final Charge	Basic Charge - Discount
Invalid Vehicle Type	The program stops if an unsupported vehicle type is entered.

IPO Table

Input	Process	Output
Entry Time	Calculate parking duration	Parking Duration
Exit Time	Round partial hours upward	Basic Charge
Vehicle Type	Determine hourly rate	Discount
Membership Status	Apply membership discount	Final Charge

Algorithm (Pseudocode)

BEGIN

INPUT EntryTime

INPUT ExitTime

INPUT VehicleType

INPUT MembershipStatus

CarRate = 3.0

MotorcycleRate = 1.5

VanRate = 5.0

MembershipDiscountRate = 0.05

// Step 1: Calculate parking duration

ParkingHours = ExitTime - EntryTime

IF ParkingHours < 0 THEN

 ParkingHours = ParkingHours + 24

END IF

ParkingHours = CEILING(ParkingHours)

// Step 2: Determine hourly rate based on vehicle type

IF VehicleType = "Car" THEN

 HourlyRate = CarRate

ELSE IF VehicleType = "Motorcycle" THEN

 HourlyRate = MotorcycleRate

ELSE IF VehicleType = "Van" THEN

 HourlyRate = VanRate

ELSE

```
    OUTPUT "Invalid vehicle type"
    STOP
END IF

// Step 3: Calculate basic parking charge
BasicCharge = ParkingHours * HourlyRate

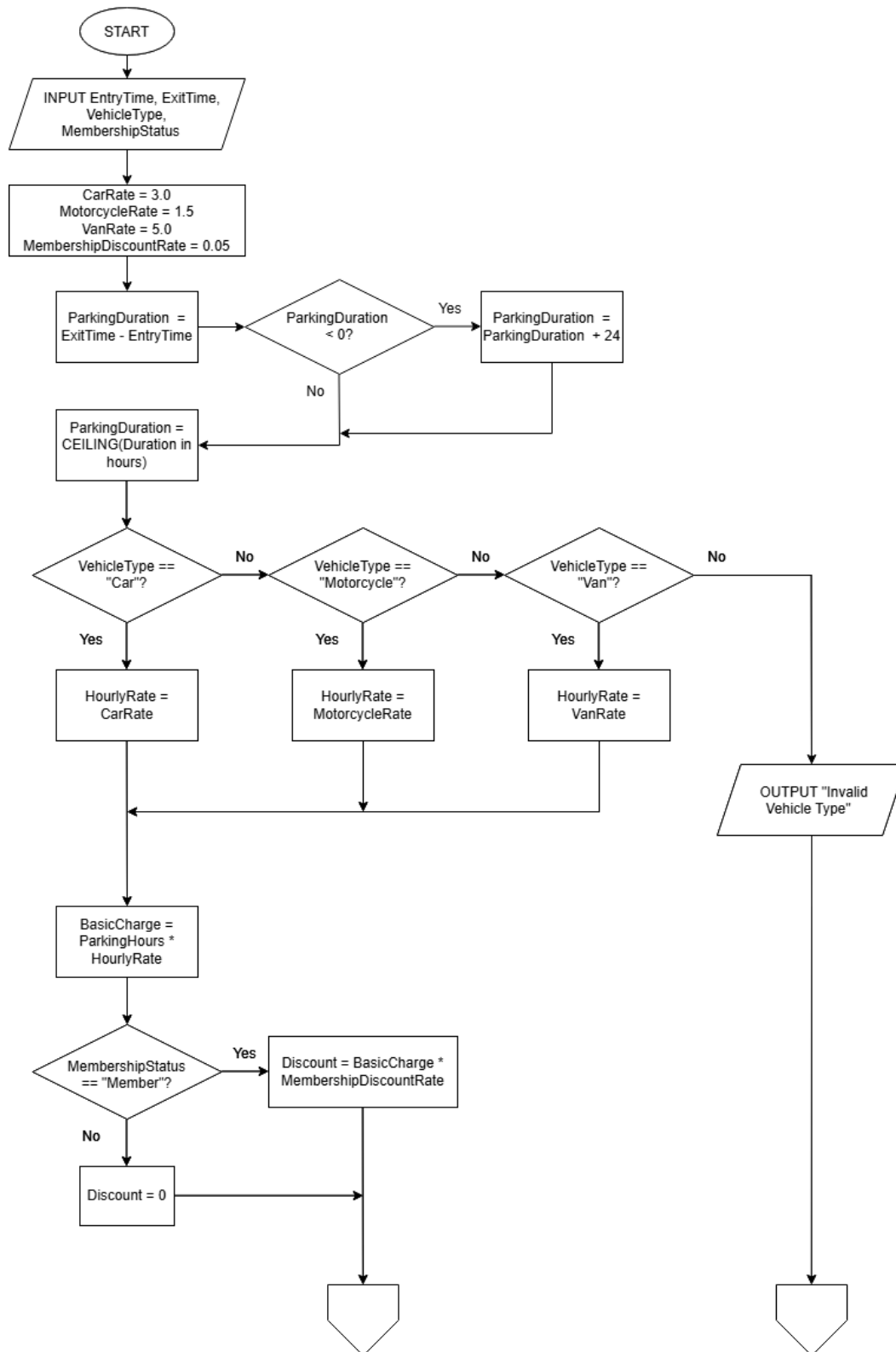
// Step 4: Apply membership discount
IF MembershipStatus = "Member" THEN
    Discount = BasicCharge * MembershipDiscountRate
ELSE
    Discount = 0
END IF

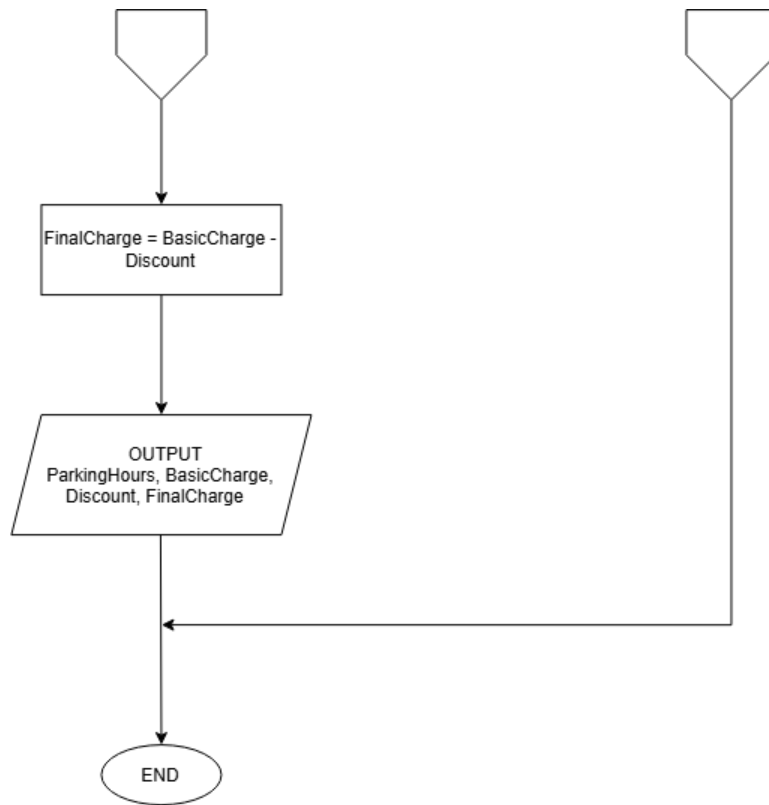
// Calculate final charge
FinalCharge = BasicCharge - Discount

// Output results
OUTPUT "Parking Duration: ", ParkingHours, " hours"
OUTPUT "Basic Charge: RM", BasicCharge
OUTPUT "Discount: RM", Discount
OUTPUT "Final Charge: RM", FinalCharge

END
```

Algorithm (Flowchart)





Three Test Cases with Expected Results

Test 1

Inputs:

Entry Time: 5

Exit Time: 7

Vehicle Type: Car

Membership Status: T (True)

Outputs:

```
Enter Time (Ex: 8 for 8:00/9.5 for 9:30 in 24H clock)

Enter Entry Time: 5
Enter Exit Time: 7

Enter Vehicle Type (Car/Motorcycle/Van): Car
Enter Membership Status (T/F): T

Parking Duration: 2 Hours
Basic Charge: RM6
Discount: RM0.3
Final Charge: RM5.7
```

Test 2

Inputs:

Entry Time: 22

Exit Time: 2

Vehicle Type: Motorcycle

Membership Status: F (False)

Outputs:

```
Enter Time (Ex: 8 for 8:00/9.5 for 9:30 in 24H clock)

Enter Entry Time: 22
Enter Exit Time: 2

Enter Vehicle Type (Car/Motorcycle/Van): Motorcycle
Enter Membership Status (T/F): F

Parking Duration: 4 Hours
Basic Charge: RM6
Discount: RM0
Final Charge: RM6
```

Test 3

Inputs:

Entry Time: 6.5

Exit Time: 10

Vehicle Type: Van

Membership Status: T (True)

Outputs:

```
Enter Time (Ex: 8 for 8:00/9.5 for 9:30 in 24H clock)

Enter Entry Time: 6.5
Enter Exit Time: 10

Enter Vehicle Type (Car/Motorcycle/Van): Van
Enter Membership Status (T/F): T

Parking Duration: 4 Hours
Basic Charge: RM20
Discount: RM1
Final Charge: RM19
```

Explanation of the Algorithm Design

This algorithm calculates the total parking fee for a vehicle based on its duration of stay, its vehicle type, and whether the driver holds an active membership.

This algorithm takes four inputs and calculates the parking fee for a vehicle based on:

- Entry time and exit time
- Type of vehicle (Car/Motorcycle/Van)
- Membership status

Inputs: Collects the entry and exit times (in hours), the vehicle classification (Car, Motorcycle, or Van), and the user's membership status (Member or Non-member).

- Constants: Initializes standard baseline pricing parameters:
- Car: RM 3.00 / hour
- Motorcycle: RM 1.50 / hour
- Van: RM 5.00 / hour
- Member Discount: 5% (\$0.05\$) off the total basic charge.

Step 1: Calculate Parking Duration

- Subtracts EntryTime from ExitTime (ExitTime - EntryTime).
- If ParkingHours < 0 (meaning the vehicle entered before midnight and exited after midnight, Ex: entered at 22:00 and left at 02:00), the algorithm adds 24 hours to normalize the negative result to the correct total duration.
- Then applies the CEILING() function to round any partial fraction of an hour up to the next full integer (e.g., a stay of 1.2 hours becomes 2 full hours).

Step 2: Determine Hourly Rate based on Vehicle Type

An if-else-if condition structure checks VehicleType to select the matching rate:

- Matches "Car" sets HourlyRate = RM 3.00
- Matches "Motorcycle" sets HourlyRate = RM 1.50
- Matches "Van" sets HourlyRate = RM 5.00

Step 3: Calculate Basic Parking Charge

Multiplies the rounded duration (ParkingHours) by the determined HourlyRate:

- $\text{BasicCharge} = \text{ParkingHours} * \text{HourlyRate}$

Step 4: Apply Membership Discount

Checks MembershipStatus:

If "Member", calculates a 5% discount:

- $\text{Discount} = \text{BasicCharge} * 0.05$

Otherwise, sets Discount = 0.

Step 5: Final Charge Calculation & Output

Computes net payable amount:

- $\text{FinalCharge} = \text{BasicCharge} - \text{Discount}$

Prints an itemized breakdown showing Parking Duration (in hours), Basic Charge (RM), Discount (RM), and Final Charge (RM).

Group Contribution Summary

Bryant: Problem analysis, assumptions and constraints, and code programming.

Gan: IPO Table, code programming, and debugging.

Farzana: Pseudocode, code programming.

Amir: Flowchart, code programming, and debugging.

Hadi: Three test cases and explanation of the algorithm design.